Heuristic Evaluation

Usability Guidelines ("Heuristics")

- Plenty to choose from
  - Nielsen's 10 principles
  - Norman's rules from Design of Everyday Things
  - Tognazzini's 16 principles
  - Shneiderman's 8 golden rules
- Help designers choose design alternatives
- Help evaluators find problems in interfaces ("heuristic evaluation")

To help relate these heuristics to what you already know, here are the high-level principles that have organized our lectures.

L = Learnability (and memorability)
S = Simplicity
V = Visibility
UC = User control & freedom
ER = Error handling
EF = Efficiency
GD = Graphic design

Nielsen Heuristics

1. Match the real world (L)
2. Consistency & standards (L)
3. Help & documentation (L)
4. User control & freedom (UC)
5. Visibility of system status (V)
6. Flexibility & efficiency (EF)
7. Error prevention (ER)
8. Recognition, not recall (ER)
9. Error reporting, diagnosis, and recovery (ER)
10. Aesthetic & minimalist design (GD)

Norman Principles

- Affordances (L)
- Natural mapping (L)
- Visibility (V)
- Feedback (V)

Tog's First Principles

minimize response time and give appropriate feedback for slow operations

1. Anticipation (EF)
2. Autonomy (UC)
3. Color blindness (GD)
4. Consistency (L)
5. Defaults (EF)
6. Efficiency (EF)
7. Explorable interfaces (UC)
8. Fitts's Law (EF)
9. Human interface objects (L)

direct manipulation: onscreen objects should be continuously perceivable, and manipulable by physical actions

10. Latency reduction (V)
11. Learnability (L)
12. Metaphors (L)
13. Protect users' work (ER)
14. Readability (GD)
15. Track state (EF)
16. Visible navigation (V)

Shneiderman's 8 Golden Rules

1. Consistency (L)
2. Shortcuts (EF)
3. Feedback (V)
4. Dialog closure (V)
5. Simple error handling (ER)
6. Reversible actions (UC)
7. Put user in control (UC)
8. Reduce short-term memory load (ER)

Heuristic Evaluation

- Performed by an expert
- Steps
  - Inspect UI thoroughly
  - Compare UI against heuristics
  - List usability problems
  - Explain & justify each problem with heuristics

How To Do Heuristic Evaluation

- Justify every problem with a heuristic
  - "Too many choices on the home page (Aesthetic & Minimalist Design)"
  - Can't just say "I don't like the colors"
- List every problem
  - Even if an interface element has multiple problems
- Go through the interface at least twice
  - Once to get the feel of the system
  - Again to focus on particular interface elements
- Don't have to limit to the 10 Nielsen heuristics
  - Nielsen's 10 heuristics are easier to compare against
  - Our 7 general principles are easier still

Shopping Cart Contents

Welcome, Ben Bitdiddle.

You have 2 item(s) in your shopping cart.
To remove an item, check "Remove" box & click "Recalculate".
Shipping Calculator below.

There is a problem with your order.

Product	Description	Quantity	UnitPrice	ExtPrice
	323022 Pinnalce Clean Plus Version 4.0 Retail ***(Free 2nd Day)*** ☐ Remove	1	$61.00	$61.00
	80098-21 Corsair VS1GBKIT400 1GB Kit DDR400 PC3200 Value Select Memory Retail (out of stock) Remove Hardware	1	$179.00	$179.00

Subtotal: $240.00

For more information about tax, please click here.

Recalculate | Clear Cart

Shipping Promotion details. Please read.

Check Out

Note: Discount will be applied during check out

Coupon Code: Apply

Ship to Zip Code: Calculate Shipping Charge

Have not made up your mind? Save all the items in your shopping cart!

Cart Title: Save Shopping Cart

Return to old shopping cart:

Cart Name: Load Shopping Cart

1. Shopping cart icon is not balanced with its background whitespace (graphic design)
2. **Good:** user is greeted by name (feedback)
3. Red is used both for help messages and for error messages (consistency, match real world)

4. “There is a problem with your order”, but no explanation or suggestions for resolution (error reporting)
5. ExtPrice and UnitPrice are strange labels (match real world)
6. Remove Hardware button inconsistent with Remove checkbox (consistency)
7. "Click here“ is unnecessary (simplicity)

8. No “Continue shopping" button (user control & freedom)
9. Recalculate is very close to Clear Cart (error prevention)
10. “Check Out” button doesn’t look like other buttons (consistency, both internal & external)
11. Uses “Cart Title” and “Cart Name” for the same concept (consistency)
12. Must recall and type in cart title to load (recognition not recall, error prevention, efficiency)

Heuristic Evaluation Is Not User Testing

- Evaluator is not the user either
  - Maybe closer to being a typical user than you are, though
- Analogy: code inspection vs. testing
- HE finds problems that UT often misses
  - Inconsistent fonts
  - Fitts's Law problems
- But UT is the gold standard for usability

Hints for Better Heuristic Evaluation

- Use multiple evaluators
  - Different evaluators find different problems
  - The more the better, but diminishing returns
  - Nielsen recommends 3-5 evaluators
- Alternate heuristic evaluation with user testing
  - Each method finds different problems
  - Heuristic evaluation is cheaper
- It's OK for observer to help evaluator
  - As long as the problem has already been noted
  - This wouldn't be OK in a user test

Formal Evaluation Process

1. Training
   - Meeting for design team & evaluators
   - Introduce application
   - Explain user population, domain, scenarios
2. Evaluation
   - Evaluators work separately
   - Generate written report, or oral comments recorded by an observer
   - Focus on generating problems, not on ranking their severity yet
   - 1-2 hours per evaluator
3. Severity Rating
   - Evaluators prioritize all problems found (not just their own)
   - Take the mean of the evaluators' ratings
4. Debriefing
   - Evaluators & design team discuss results, brainstorm solutions

Severity Ratings

- Contributing factors
  - Frequency: how common?
  - Impact: how hard to overcome?
  - Persistence: how often to overcome?
- Severity scale
  1. Cosmetic: need not be fixed
  2. Minor: needs fixing but low priority
  3. Major: needs fixing and high priority
  4. Catastrophic: imperative to fix

Evaluating Prototypes

- Heuristic evaluation works on:
  - Sketches
  - Paper prototypes
  - Buggy implementations
- "Missing-element" problems are harder to find on sketches
  - Because you're not actually using the interface, you aren't blocked by feature's absence
  - Look harder for them

Writing Good Heuristic Evaluations

- Heuristic evaluations must communicate well to developers and managers
- Include positive comments as well as criticisms
  - "Good: Toolbar icons are simple, with good contrast and few colors (minimalist design)"
- Be tactful
  - Not: "the menu organization is a complete mess"
  - Better: "menus are not organized by function"
- Be specific
  - Not: "text is unreadable"
  - Better: "text is too small, and has poor contrast (black text on dark green background)"

Suggested Report Format

- What to include:
  - Problem
  - Heuristic
  - Description
  - Severity
  - Recommendation (if any)
  - Screenshot (if helpful)

Under «Heuristic Evaluation» section

12. Severe: **User may close window without saving data** (error prevention)

If the user has made changes without saving, and then closes the window using the Close button, rather than File >> Exit, no confirmation dialog appears.

Recommendation: show a confirmation dialog or save automatically

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Cognitive Walkthrough: Another Inspection Technique

- Cognitive walkthrough = expert inspection focused on learnability
- Inputs:
  - prototype
  - task
  - sequence of actions to do the task in the prototype
  - user analysis
- For each action, evaluator asks:
  - will user know what subgoal they want to achieve?
  - will user find the action in the interface?
  - will user recognize that it accomplishes the subgoal?
  - will user understand the feedback of the action?

Summary

- Heuristic evaluation finds usability problems by inspection